Enrollment No.: _____

Darshan UNIVERSITY

Darshan Institute of Computer Application
BCA / B.Sc. (IT) | Sem-2 | Summer-2025

Course Code	:	2304CS205	**Date**	:	21-04-2025
Course Name	:	Operating System	**Duration**	:	150 Minutes
			Total Marks	:	70

Instructions:
1. Attempt all the questions.
2. Figures to the right indicates maximum marks.
3. Make suitable assumptions wherever necessary.

Q.1 **(A)** Define Operating System and discuss functions of Operating System **4**

(B) Discuss Time-Sharing Operating System with proper example **3**

OR

Explain Batch Operating System with suitable example.

(C) Discuss services of Operating System **7**

OR

Define System Call. Explain types of System Calls.

Q.2 **(A)** Explain Process States Transitions with a suitable diagram. **4**

(B) Discuss Context Switching with proper example **3**

OR

Define Thread. State the difference between Thread and Process.

(C) Solve following example by FCFS and SJF cpu scheduling algorithm. Draw gantt chart and calculate average waiting time and average turnaround time. **7**

Process	Arrival Time	Burst Time
P0	0	5
P1	1	3
P2	2	8
P3	3	6

OR

Consider four processes: P1, P2, P3, and P4, with an arrival time of 0 ms, 1 ms, 3 ms and 5 ms respectively. Their respective burst times are 6 ms, 5 ms, 2 ms, and 1 ms. calculate the finish time, average waiting time and turnaround time using SRTN and Round Robin (quantum time = 2ms) scheduling.

Q.3 **(A)** Define Deadlock. Explain characteristics of deadlock. **4**

(B) Define the following terms. **3**
Race Condition, Critical Section, and Mutual Exclusion

OR

Discuss Deadlock Detection and Recovery with suitable example.

(C) List out the mechanisms for achieving mutual exclusion with busy waiting and explain Disabling Interrupts with a suitable example. **7**

OR

List out the mechanisms for achieving mutual exclusion with busy waiting and explain Shared Lock Variable with a suitable example.

Q.4 **(A)** Consider six memory partitions of size 200 KB, 400 KB, 600 KB, 500 KB, 300 KB and 250 KB. These partitions need to be allocated to four processes of sizes 357 KB, 210 KB, 468 KB and 491 KB in that order. Perform the allocation of processes using first fit, next fit, best fit, and worst fit memory allocation scheme. **4**

(B) State the difference between Logical Address and Physical Address. **3**

OR

Explain variable (dynamic) partition memory allocation technique with a suitable example.

(C) Consider the page reference string 1, 2, 3, 4, 1, 2, 5, 1, 2, 3 with 3-page frames. Calculate the number of page faults using FIFO and LRU Page Replacement Algorithm. **7**

OR

Consider the page reference string 4, 5, 6, 7, 4, 5, 8, 9, 4, 6, 7 with 3-page frames. Calculate the number of page faults using FIFO and Optimal Page Replacement Algorithm.

Q.5 **(A)** List operations perform on files and discuss all in brief. **4**

(B) Define the file. Discuss different types of files in brief. **3**

OR

Define I/O buffering. Explain types of I/O buffering in brief.

(C) Consider an imaginary disk with 200 cylinders (numbered 0 to 199). A request comes in to read a block on cylinder 51. While the seek to cylinder 51 is in progress, new requests come in for cylinders 98, 183, 37, 122, 14, 124, 65, and 67 in that order. Starting from the current head position, what is the total distance (in cylinders) that the disk arm moves to satisfy all the pending requests, for each of the following disk scheduling Algorithms? **7**

I. FCFS/FIFO

II. SSTF

OR

Consider an imaginary disk with 51 cylinders (numbered 0 to 50). A request comes in to read a block on cylinder 11. While the seek to cylinder 11 is in progress, new requests come in for cylinders 1, 36, 16, 34, 9, and 12 in that order. Starting from the current head position, what is the total distance (in cylinders) that the disk arm moves to satisfy all the pending requests, for each of the following disk scheduling Algorithms?

I. SCAN

II. C-SCAN

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